# Mahmoud Bekri, PhD, CIB

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## BACKGROUND

Mahmoud Bekri currently serves as a Research Economist at the Islamic Development Bank Institute (IsDBI). He holds a PhD in Economics and Finance and a German Diploma in Physics (equivalent to a bachelor's and a master's degree combined), both from the Karlsruhe Institute of Technology (KIT) in Germany, where he subsequently worked as a Postdoctoral Researcher at the Institute for Economic Theory and Statistics.

With a strong foundation in Mathematics and Physics, Mahmoud Bekri's advanced Quantitative and Analytical skills deepen his expertise in economics and finance. His career combines Academic and Professional pursuits; he taught Advanced courses in Financial Engineering, has contributed to Scientific Research in Islamic, Ethical, and Social Finance. Additionally, he has provided Consultancy on developing Advanced Financial Methodologies for International Banks and Asset Management Firms. His previous roles also include Strategist, Project Manager, and Sales Manager. He also regularly publishes in peer-reviewed scientific journals.

Since joining the IsDB Group in 2015 as part of the "Young Professionals" Program, Mahmoud Bekri has played a key role in numerous projects and led Global Initiatives to advance the Islamic Finance Sector across various departments, including the Islamic Finance Services Department, Financial Product Development Center, IRTI, and, most recently, the IsDB Institute. He has earned the Certified Islamic Banker (CIB) designation in 2016, the Leadership Certificate in Islamic Banking and Finance in 2018, and the Certificate in Advanced Project Management in 2024.

## EDUCATION

2024 **Certificate in Advanced Project Management**
Meirc Professional Certificate (MPC) member of PMI *financed by employer*

2018 **Leadership Certificate in Islamic Banking and Finance**
Cambridge Islamic Finance Leadership (IFLP) Programme 2018 *financed by employer*

2016 **Certified Islamic Banker (CIB)** of *CIBAFI financed by employer*

2014 **PhD** Karlsruhe Institute of Technology (KIT), Germany
Institute of Economics (ECON), Chair of Econometrics and Statistics
*Title:* **"Portfolio Optimization and Risk Management under Non-normal Stable and Tempered Stable Distributions in Islamic Finance"**
*obtained magna cum laude*

**Supervision**: ***Prof. Rachev*** *Stony Brook US*, ***Prof. Heller*** *KIT Germany*,
***Prof. Fabozzi*** *Yale University US*

2011 ***German Diplom*** **Physicist** Karlsruhe Institute of Technology (KIT), Germany
**Diploma thesis:** „Studies of Si-MCM-41 and Al-MCM-41 Surface Structure and Surface Charge" (Supervision of Prof. Schimmel)
*obtained with final grade "very good" (sehr gut)*

## RELEVANT PROFESSIONAL EXPERIENCE

01/2026 - Present **Secondment, Model Risk | Risk Management Department, IsDB**

- Contribute to building the Bank's model risk management framework.
- Designed a multi-manager equity risk oversight framework that monitors external equity managers against their benchmark (tracking error, risk attribution and performance assessment), documented for formal model validation.
- Built an asset-liability management cash-flow stress model and an Islamic ALM model, with full reconciliation and quality-control checks.
- Supported hedging analysis, including swap hedge accounting, and delivered an internal workshop on hedging Islamic equity portfolios.

2018 - Present **Research Economist | Islamic Development Bank Institute (IsDBI)**

- Lead research, policy advisory, and capacity-building initiatives to advance Islamic finance across IsDB member countries.
- Conduct applied economic and quantitative research on sustainable finance, risk-sharing instruments, and fintech innovation.
- Develop analytical frameworks and financial models supporting the design of Islamic financial products and sovereign instruments such as Sukuk.
- Collaborate with central banks, ministries of finance, and development partners to strengthen Islamic finance ecosystems and enhance regulatory capacity.
- Represent the Institute in international forums and conferences, promoting thought leadership on Islamic finance, digital economy, and sustainability.
- Supervise and contribute to publications, working papers, and knowledge products advancing the Institute's research agenda.

**Notable Achievements**

- Designed and delivered high-impact capacity-building programs for central banks and financial regulators in Libya, Morocco, Uganda, and Mauritania.
- Contributed to the development and approval of several Islamic Finance Grants supporting Awqaf, participatory finance, and social finance initiatives across member countries.
- Co-authored pioneering research on GDP-linked Sukuk, Islamic financial benchmarks, and inflation modeling post-COVID-19.
- Played a key role in IsDBI's patent-winning "Smart Voucher System" (US Patent #12282916), applying fintech innovation to development finance.
- Published multiple peer-reviewed papers and thought leadership blogs on Islamic finance, sustainable development, and financial innovation.
- Authored the book *"Heavy Tails and Coherent Risk Measures in Islamic Finance"* on advanced financial modeling techniques.
- Delivered the Capacity Building Program for Interuniversity Postgraduate Program in Economics (PTCI), in Dakar, Senegal on "Introduction to Islamic Finance"
- Built the database and performed the quantitative analysis on the Consensus Economics Survey Data, used to examine the odds of building an Islamic Financial Returns Benchmark for Islamic Financial Institutions
- Authored "Inflation Dependence Structure in a post-COVID-19 World" and "Crypto GDP Sukuk, for Better Debt Servicing in Post-COVID-19" Blogs and translated 3 blogs from English into Arabic
- Drafted and finalized the IsDBI contribution to IsDBG-MCPS for 3 MCs: Niger, Guinea and Morocco
- Co-authored the paper "Taming Inflation: An Islamic-finance Perspective"

- Representative of IsDBI in Libya Task Force, delivering trainings to Libyan financial institutions: Central Bank of Libya, Libya Development Bank,...
- Coordinator of the Palestine Monetary Authority Digital Currency Project
- Member of the organization team of the first prominent Economists' Forum under the theme "The Rise of Central Bank Digital Currency: Motives, Approaches, and Techniques".
- Contributed to translating IsDBI works from English into French and Arabic
- Nominated Technical Assistance Project officer for Projects in Uganda, Burkina Faso and UAE
- Authored and presented a manuscript "On the role of cryptocurrencies in Islamic Finance, real economy and sustainability" in Istanbul, Turkey
- Led a Task Force and performed the quantitative work on "Developing an Islamic Financial Returns Benchmark for Islamic Financial Institutions"
- Contributed to obtaining two patents "The Blockchain based Voucher System" & "Blockchain based Credit management"
- Authored & translated several blogs on a range of contemporary issues in economics & finance, Fintech, sustainable development & related matters
- Designed a fintech product to support sovereign Islamic finance initiatives
- Worked on the Smart Economy Project to patent and implement a fintech product
- Presenter and Panelist in ADFIMI -Qatar Development Bank Joint Workshop on "Sharia Compliance Issues of Crypto Assets" in Qatar
- Member of the team on Center of Excellence on Islamic Finance: Successful organisation of the first workshop in Madinah
- Submitted the research paper "The Determination of Alpha in CAR "to the Tunisia International Conference on IF (TICIEF 2019)
- Conducted an international workshop to address the challenges of capital regulations for Islamic financial institutions, in Sultanate of Oman
- Evaluator of the Queen Alia Airport Project in Amman, Jordan. Joint mission with the Operations Evaluation Department (OED) of IDB
- Pricing Analyst and Product developer, member of the team on the design of the Sukuk Assets Utility Awqaf
- Authored *"The Determination of Alpha in the Capital Adequacy Ratio for Islamic Financial Institutions"*
- Authored a case study of the Islamic Banking Africa Summit as an advocacy platform
- *Co-Editor of the Islamic Finance Country Report on Djibouti*

2015-2018 **Young Professional | Islamic Development Bank Institute (IsDBI)**

*The Young Professionals Program (YPP) is the strategic talent pipeline for the professional career in the IDB Group. The program comprises a training program and professional program (hands-on rotations), described below:*

***Financial Product Development Center, 3rd rotation***

- Represent the department in the Risk Management Committee of IDB Group
- Author the "*Survey, Study & Analysis of the Financial Structures Depending on GDP Shares, Commodities Linked-Shares & Similar Structures*"
- Author the 1st part of the study on: "*Fiduciary and Third-Party Assessment Risk*"
- Participate in the design of Sovereign Finance Company (SFC) and its Cash Flow Model
- Participating in the Development Effectiveness Review of the IsDB
- Analyze the Corporate and US Treasury Bonds used to price Sukuk.

***Financial Product Development Center, 2nd rotation***

- Represent the Department in the Risk Management Committee of IDB Group
- Participated in the 29th International Summer School of the Swiss Association on the Actuaries on Quantitative Risk Management in Lausanne

- Author a "*Survey of the CDS and a brief Survey of the Modeling of CDS*"
- Determine the Premiums of Performance Sensitive Financial Papers
- Author a First Draft of an Export Index relevant to the OIC Countries' Exports
- Optimize the Stocks' Basket of Exchangeable Sukuks picked up by SocGen
- Build a Sukuk Databank and analyze the tail Risk of tradable Sukuk

***Islamic Financial Services Department (IFSD), 1st rotation***

- Audit the Saudi Project for Hajj Meat (**Adahi**) managed by IDB
- Present the Zitouna Tamkeen Project to the Investment Committees and prepare the Due Diligence Questionnaire and the Final Project Report
- Contribute to the Technical Assistance (TA) Process Review: contribute to the Team Meetings and in drafting the TA Request Form
- Review the Feasibility Study from Financial and Legal Points of View of the Awqaf Properties Investment Fund Project in Cotonou Benin, participate in the Appraisal Mission and the Meetings with the consultants and the ministry of finance and contribute to the refinement of the costing of the project
- Contribute to the Islamic Financial Services Board Council Meeting
- Prepare the Presentation about the IF Gates in Kuwait and Luxembourg
- Contribute to the Committee for Establishing the Corporate Identity and Marketing of the IFS Department Activities

11/2013-02/2015 **Postdoctoral Researcher**
***Economic Theory and Statistics Institute (ETS) KIT Germany***

- Develop the Scientific Work on Financial Engineering, Portfolio Optimization and Risk Management
- Manage the Research Projects on Islamic Finance
- Supervise Bachelor and Master theses and co-supervise a PhD thesis
- Deliver Courses on Financial Engineering and advanced Financial methods
- Deliver Seminars on Hedging and Financial products

11/2013-09/2014 **Senior Consultant in charge of IF projects**
***FinAnalytica & Bravo Risk Management NY***

- Develop Investment Portfolio Analytics and Assets Management Tools
- Impact long-term Investment Portfolios' Performance with tail Risk Analytics
- Optimize Investment Portfolios: Enhancing Performance and mitigating Risk
- Apply Stable Distributions on the Components of the German Dax index
- Apply heavy-tail Distributions on Sukuks
- Develop Risk Management Software tool for daily trading in IF

04/2013-10/2013 **Senior Social Network (SNA) Analyst & Consultant**
***SONEAN GmbH (powered by Funds@Work) Frankfurt, Germany***

- Develop SNA based Business Strategies
- Assist and make the follow-up of the implemented Strategies
- Create and continuously extend the Database of SONEAN
- Analyze systematically the Social Capital and Networks of decision makers
- Identify Business Risks and Opportunities for clients

12/2012-02/2013 **Sales Manager at *Edmund Optics (EO) Karlsruhe, Germany***
***Leading Producer of Optics, Imaging, and Photonics Technology***

- Provide Customer-and Application-oriented Technical Advice
- Perform the Feasibility Analysis and calculate the Offers

- Sell the customized Products and process the Delivery Requests

09/2012-11/2012 **Internship in Portfolio Pricing Department**
***Hoist Finance Group Duesseldorf, Germany***

- Gain an Overview of the various Methods and Challenges of the Rating/Assessment of unsecured Non-Performing Loans (NPL)
- Gain Knowledge of the Different Phases of Debt/Receivables' Management
- Price NPL Portfolios and determine Recovery and Cost curves
- Create and Present the template of the performed Portfolio Assessment for the Investment Committee on the Portfolio Purchase

03/2011-08/2012 **Project Manager at *TISFO***
**(*Tunisian Industrial Services for Oil*) *Sousse, Tunisia***

- Effectuate the Market Research and Competitive Analysis
- Manage the Progress of the Realization of Projects: Planning, Implementation, and final Control
- Strengthen the Business Relations with previous and new Customers
- Coordinate with the different Departments and Stakeholders to meet the usually non flexible Deadlines
- Control the Budget

## REFEREE FOR

- 2015 International Journal of Islamic and Middle Eastern Finance and Management
- 2014 Frontiers in Mathematical Finance

## SELECTED PUBLICATIONS

**Thesis Published**

- **Bekri, M.** (2014) 'Portfolio Optimization and Risk Management under Non-normal Stable and Tempered Stable Distributions in Islamic Finance'. Doctoral thesis, KIT, Germany

**Book and Chapter in Book**

- Book: (2021) **Bekri, M.** *Heavy Tails and Coherent Risk Measures in Islamic Finance*. Islamic Development Bank Institute, Jeddah. 170 pp. ISBN 978-603-8306-06-2
- Book Chapter: (2017) "On the dependency structure of Islamic assets" **Bekri, M**., Kabir Hassan, M., and Alam, N., Handbook of Empirical Research on Islam and Economic Life. Edited by M. Kabir Hassan. pp 602–623

**Indexed Scientific Journals**

- Hajjar, M., **Bekri, M.** (2026). Quick Commerce and Islamic Ethics: Maqāṣid and Policy Assessment. *International Journal of Islamic Finance and Sustainable Development*, 18(3), 41-59. doi:10.55188/ijifsd.v18i3.1461
- Mirzaei, M.**, Bekri, M**. (2017). Energy consumption and CO2 emissions in Iran, 2025. *Environmental Research*, 154, 345-351. (ISI, IF: 4.2)

- Mirzaei, M., **Bekri, M**., Ramakrishnan, S., (2016). Corporate Default Prediction with Industry Effects: Evidence from emerging markets, *International Journal of Economics and Financial Issues, 6(S3), 161-169*

- Ramakrishnan, S., Mirzaei, M., **Bekri, M**. (2015). Corporate Default Prediction with Adaboost and Bagging Classifiers. *Jurnal Teknologi,* 73(2), 45-50

- Ramakrishnan, S., Mirzaei, M.**, Bekri, M.** (2015). A Multi-Industry Default Prediction Model using Logistic Regression and Decision Tree, *Research Journal of Applied Sciences, Engineering and Technology*, 9(10), 856-861

- Ramakrishnan, S., Mirzaei, M., **Bekri, M.** (2015). Adaboost Ensemble Classifiers for Corporate Default Prediction. *Research Journal of Applied Sciences, Engineering and Technology*, 9(3). 224-230

- **Bekri, M.**, Kim, Y. S. (2015) 'Tail risk analysis of the S&P/OIC COMCEC 50 index'. *Borsa Istanbul Review* 15(1), 1–16

- **Bekri, M.**, Kim, Y. S. (2014) 'Portfolio management with heavy tailed distributions in Islamic Finance'. Journal of Islamic Economics, Banking and Finance (JIEBF), 10(2), 114-137

- **Bekri, M.**, Kim, Y. S., Rachev, S. T. (2014) 'Tempered stable models for Islamic finance asset Management'. International Journal of Islamic and Middle Eastern Finance and Management, 7(1), 37-60

## LANGUAGES

1. Arabic : Mother tongue
2. English : Fluent
3. German: Fluent
4. French: Fluent

## SOFTWARE

MS Office (Word, Excel, PowerPoint, Access) (good); MATLAB and Python (frequent user).

## INTERESTS AND HOBBIES

Reading, Writing, Sport and Traveling

## OTHER ACTIVITIES

- Member of the German Business Circle Meeting of the German-Saudi Arabian Liaison Office for Economic in Jeddah, Saudi Arabia
- Member of the Tunisian Scientific Society (TWG) in Karlsruhe in Germany
- Member of the Academic "Wednesday Talk": Meeting of the Economic Scholars in King Abdulaziz University in Jeddah, Saudi Arabia
- Former Chairman of the Jury of the Master of Islamic Financial Management in Effat College of Business in Jeddah, Saudi Arabia